

SDS 210 - Introduction to Statistics

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1. WELCOME TO SDS 210

I am so excited that you'll be joining this educational journey through statistics! You are now, just by being here, a statistician. This course will help us frame out exactly what that means and also what *statistics* is.

Note

You are now a statistician!

1.a. Official Course Description

An application-oriented introduction to statistical modeling, covering topics of descriptive statistics, data visualization, point and interval estimates, bivariate and multiple regression modeling, and inferential hypothesis tests using both distributional and resampling methods. Lectures include “hands on” demonstrations of statistical phenomenon, with labs and assignments that emphasize analysis of real data.

1.a.i. Motivating Questions:

There are a few questions motivating our course work:

- What is Statistics?
- What role does data play in statistics?
- What habits of mind do we need to develop to be effective and ethical statisticians?

1.a.ii. Course Learning Objectives:

By the end of the course, statisticians (ie. you!) will be able to...

1. Responsibly use sample data to learn about a larger population, using exploratory and inferential statistics - *Create Statistics*
2. Detail the power and limitations of statistical concepts, and use this understanding to evaluate statistics from various contexts - *Evaluate Statistics*
3. Explain the connection between probability and inferential statistics - *Connect Statistical Concepts*
4. Work collaboratively, responsibly, and reflectively to create and present statistics to a wide variety of audiences - *Communicate with Statistics*

1.b. Syllabus

This webpage and the associated links together function as the syllabus. Everything that you need to know about the course is either here or linked from here.

The course syllabus is the most important document underlying the culture of our course and our classroom. I view my syllabus as a sacred document that both introduces and governs the course. In taking this view, I work to detail as much about the course as possible from the big picture ideas to the minute details of course policies. I believe that the first activity a student should do when beginning a course is to carefully read and examine the syllabus. In keeping with that belief, there are a few tasks in this document that will help us build our course community.

1.c. Course Acknowledgements

Parts of this course - including slides, activities, and notes - will be from a variety of sources. Materials will be appropriately attributed and will be used in keeping with copyright and fair use laws.

This course is a version of DataSci 220, *Data-Driven Statistics with AI* taught at the US Air Force Academy. I was part of the original development and teaching team for that course in Spring 2025. I continue to be in collaboration with the Course Director there, and we will be running similar versions of our courses at the same time with adjustments appropriate for each school.

2. COURSE PHILOSOPHY

This course is designed to be a first course in Statistics for students with **no prior experience** in or with statistics. This course seeks to begin your training in statistics. Success in this course will be measured iteratively, placing emphasis on the student's ownership of the learning process and a student's consistency of effort.

Note

If you have prior experience with Statistics, please reach out to the instructor. We want to make sure that you are in the right course *for you*.

2.a. Stretch Zone

Borrowing language from the 2019 Google J-term course, when learning in a classroom setting, at any moment, we exist in one of three zones: safe, stretch, or strain. A student in the safe zone is completely comfortable, and the course material is completely within what the student already knows. The strain zone is the complete opposite of the safe zone. A student in the strain zone feels completely overwhelmed with the material to the point where they can't find any meaningful connections between the current material and what the student already knows.

The material in this course is intellectually challenging. Several of the topics are ones that took me a long time to fully understand their subtleties. The goal for this course is to spend the vast majority of our time in the stretch zone. In this zone, the material is just beyond our current knowledge base and while we are a bit uncomfortable with the material, we can see possible paths to tie the new material into our existing knowledge. In other words, in this class, our brains should be stretching, but not straining.

At the end of each week, if you have not been surprised or challenged, or if you have no questions about the material, this could mean that you are in the safe or strain zones. If you find yourself exasperated or negative about the class, you may be in the strain zone. If you find yourself using words like "boring" or "easy", you are likely in the safe zone. If you are in either zone, please get in touch with me: in class, on slack, in an appointment, or send me an email. I want you to get the most out of this course, but I cannot make adjustments to the course if I do not have firsthand knowledge of what is going on.

3. PROFESSIONAL DEVELOPMENT

As we explore statistics, we will also be developing as professional statisticians. Specifically, we will focus on what it means to be a member of a professional environment and how to be intentional and reflective in our work.

3.a. *Participating in the Course*

Being part of a professional community has many parts; we generate and consume knowledge, we review and critique work, and we present to and learn from each other. The edge of what we know is constantly being pushed further and further as we discover new ideas, unearth new techniques, and ask new questions. At times, we are programmers, writers, and gardeners, and then at other times, we serve as reviewers, mentors, and learners. The constancy is our willingness to be present and active within our professional community. With this in mind, in this course, your intentional participation is **expected**.

Participation can take many forms, including but not limited to: attending class meetings, being active on our course slack, participating in discussion in class, offering ideas and questions in class or on slack. As a community our class will generate a list on our google drive of what we believe are ways to engage and participate in our course.

Note

Our class's definition for participation can be found [here](#)

Adding to this list one way that you are called to participate in this course. If you find yourself unwilling or resistant to add to this list, examine this. What is holding you back? What would help propel you to action?

3.b. *Preparing for Class*

Participating our course will require more than being physically present; it is being prepared to engage with the topics, concepts, and activities of the day. This means having completed the reading, and any other pre-class activities for the day.

Completion in this context means giving yourself the time and space to finish the most recent reading (asking questions on slack as you have them), to formulate half-formed questions and comments about the readings, and to try, fail, and try again on the group problems. In this way, completion is more about expanding our ideas (by coming up with questions and comments) about machine learning, than it is about perfectly finishing all the things. This kind of preparation means:

- Blocking time in your daily work life for this course
- Noting and sharing each question you have
- Reading with a pen and highlighter in your hand
- Taking frequent notes on readings, discussions, and labs
- Working on pieces of the group problems and reviewing what your group discussed

- Gathering the fragments of answers to the questions that you and others have posed

3.c. *Intentional and Reflective Work*

A project involving statistical procedures is rarely a straightforward pursuit with a clear path from beginning to end. I have found the most success when I engage a process of constantly reflecting on where I am in a project compared to where I began and then intentionally determining my next step based on those assessments. In this class, we will reflect frequently. First we will mark the beginning and the end of the semester with a general check-in on where we are in our educational journeys.

3.d. *Writing*

This course focuses on statistics and while it might be tempting to assume that “statistics” means “complicated and inaccessible writing”, this is not true for our course. For each assignment, you will be asked to explain your process, detail how the various pieces interact, and illuminate any potential weaknesses. This is excellent training for the ‘real world’ where you will have managers, users, and customers with varying levels of technical expertise.

In your assignments, I expect that you treat your readers as colleagues who have your respect and who you are genuinely interested in sharing your ideas with. This means that you may need to edit your responses or rewrite a solution to a problem for additional clarity. If you find yourself writing “the thing” or “you know what I mean,” I would recommend taking that as a sign that your assignment needs additional editing. If you have any questions about writing for statistics or more general, you are always welcome to reach out to me.

3.d.i. *Writing Enriched Curriculum:*

The project will also scaffold explicit development of writing skills that SDS has laid out in our department’s newly adopted [writing plan](#). Specifically, our course will focus on three of the listed writing goals:

- Writing Goal 2 - Engage in a writing process that includes brainstorming, outlining, initial drafting, peer review, editing, and revising;
- Writing Goal 4 - Prioritize the important parts of the process and/or project to communicate
- Writing Goal 5 - Clearly communicating the research question and how analysis will support the question in the introduction.

3.d.ii. *Jacobson Center for Writing, Teaching & Learning:*

Smith has an additional resource for writing support: the Jacobson Center for Writing, Teaching & Learning make an appointment to take your work to the Jacobson Center on [their website](#).

4. COURSE LOGISTICS

4.a. *Class Meeting Times*

The course meets three times per week on Mondays from 1:40 PM to 2:55 PM, and then Wednesdays and Fridays from 1:20 PM to 2:35 PM in McConnell 102. In supporting our course community, students should make every effort to be at our course meetings on time.

4.a.i. *Attending Lecture:*

Smith College is a residential 4-year college, and as such, this class is designed to leverage our time together. However, you do not need to email me to ask permission to not come to class. That being said, if you are not able to be in-person for 2 consecutive meetings, then we need to check in.

Similarly, do not come to student hours nor appointments if you are ill. Please ask for a zoom link to attend student hours virtually, and a zoom link can be provided for any in-person appointment.

Warning

Do **NOT** come to in-person class or student hours, if you are ill.

Failure to respect this policy will result in an email to both the class dean and your advisor.

4.b. *Low(er) Tech Class*

This class is designed to be a “low(er) tech” class. This means that we will often have no computers open or out. The instructor will signal when we need our computers. The purpose of this design is to support our learning by making clear when we need and when we do not need technology. Additionally, this version of the class is emphasizing crafting strong mental models for the material, which we then supplement with work in R.

4.b.i. *Generative AI:*

In general, you should not need to use generative AI technologies in this class. That being said, this technology exists and is becoming increasingly embedded in other systems that we use. If you find yourself reaching for a LLM or other genAI tools, please **document** your use with a five part documentation statement:

1. What system did you use and which version?
2. What prompt did you give it?
3. What did you get back?
4. What did you change before submitting the assignment.
5. Include a PDF of your full transcript.

4.c. *Communication*

In addition to our synchronous meetings, our class will make of electronic communication, including our class slack, email, and Moodle messages. These methods of communication represent differing levels of formality and collaboration:

- **Slack:** Our slack site is the primary form of course communication. It allows for us to share where we are stuck with a reading, idea, or assignment, and where we can add helpful hints, request study groups, or share interesting news articles. Slack is much less formal than email, and salutations and signatures are not required. If you need to ask your instructor a question, this is the best place to do it either over a public channel or through direct message.
- **Email:** Email communication is more formal than slack. It should be used in the cases that 1) concern something personal, 2) require attachments, or 3) involve a number of people (who should be copied on the email). We also will use email to confirm individual appointments. Emails to the instructor should include a salutation and a signature.
- **Slack Mentions:** When I need to communicate with the whole class on a time-sensitive manner, I will use the @everybody or @channel functionality in slack. Please check that your slack settings will notify you either via slack or email. If you are using the email notifications, please be sure that your email does not treat these messages as spam.
- **Slack Emojis:** At times, I will ask you to emoji a message to show that you have read the message. This is a quick way for you to signal that you read it. If there's something that you have a question about in a "please emoji" message, please use the threading function for that message or send me a direct message on slack.

Note

Please sign up for our slack site (see Moodle for the link!), add a profile picture, and introduce yourself with your name and a fun-fact about yourself on the #intros channel. After signing up, please determine how and where you will get notifications from slack.

Please note that on a typical workday, I leave my office just after 4pm. This means that emails and slack messages sent after 4pm will likely be received on the next business day. Similarly e-communication sent on the weekend will likely not be received until Monday.

4.d. Books

The readings from this course will come from three freely available digital textbooks as well as other digital readings supplied by the instructor:

1. *OpenIntro Statistics*, 4th edition by David Diez, Mine Çetinkaya-Rundel, and Christopher D. Barr. It is part of the OpenIntro project and as such is [freely available](#) as a PDF or on the web.

2. *Introduction to Modern Statistics*, 2nd edition by Mine Çetinkaya-Rundel and Johanna Hardin. It is part of the OpenIntro project and as such is [freely available](#) as a PDF or on the web.
3. *Modern Dive*, 2nd edition by Chester Ismay, Albert Y. Kim, and Arturo Valdivia. The free e-book is available [here](#)

4.e. *Instructor Information*

The instructor for this course is Katherine M. Kinnaird, Ph.D. and I use she/her pronouns. My go-by for teaching is Prof. Kinnaird (or Prof. K.). My office is room 108 Bass Hall. The **best way** to reach me is on **slack**. If you need to email me, my email is kkinnaird@smith.edu.

Note

To book an appointment with me, go to my [appointment calendar](#)

5. ASSIGNMENTS

To prepare students to tackle the course's motivating questions - especially *What habits of mind do we need to develop to be effective and ethical statisticians?* - there are a variety of assignments in this course. Each week will be a mix of reading about topics in statistics, discussing and debating concepts, putting our readings into practice, and constant reflection about our learning and professional development. Due dates are posted on the Detailed Course Schedule and are due on the stated dates at 10 am.

Note

The Detailed Course Schedule (link on moodle!) lists the due dates for the course. To view it, please log into your **Smith** account.

5.a. Engagement & Class Preparation

For each lecture meeting and lab, there will be readings and/or a pre-class activity. These readings and activities are intended to be your first contact with the course material. Engaging with these materials in good faith *before* class will help you get the most out of the in-class activities.

5.b. Homework

The course is divided into 4 blocks, and there are 2 homework assignments for each block. Each homework assignment will have two parts: Concept Checks and Core Competencies.

- The Concept Checks are online homeworks submitted on Gradescope and auto-graded. Questions become available after each lesson.
- The Core Competencies are hand-written and then submitted on Gradescope. These are graded by a human and are in the same style as questions that will appear on the Graded Reviews (GRs).

5.c. Projects

There will be one group project for the semester. In this project, you will use what we have learned in this class to investigate one data set of your choosing. There will be lab time set aside to work on the project.

The project will also scaffold explicit development of writing skills that SDS has laid out in [our writing plan](#). Specifically, our course will focus on three of the listed writing goals:

- Writing Goal 2 - Engage in a writing process that includes brainstorming, outlining, initial drafting, peer review, editing, and revising;
- Writing Goal 4 - Prioritize the important parts of the process and/or project to communicate
- Writing Goal 5 - Clearly communicating the research question and how analysis will support the question in the introduction.

5.d. Graded Reviews

There are 5 Graded Reviews (GRs)–4 during the semester and one during finals period–during which you can collect statistics tokens. There are 12 statistics tokens, each representing core skills for the course. During each GR, you can attempt as many or as few tokens as you would like. Your goal, by the end of the semester, is to collect all 12 tokens.

The tokens are available on the following pacing:

1. GR1: Tokens 1 - 2
2. GR2: Tokens 1 - 5
3. GR3: Tokens 1 - 9
4. GR4: Tokens 1 - 12

Final GR: Tokens 1 - 12

You do not need to (nor should you) attempt all tokens on each GR, but you do need to collect all 12 tokens by the end of the term.

Note

The Core Competency Homework Questions will be in the style of the GR questions.

5.d.i. *Statistics tokens:*

The 12 tokens that represent core skills for the course are listed below, in the order that they appear in the course:

Standards for Block 1 – Probability Models:

1. Compute and interpret probabilities within a given context; Construct and analyze probability tables; State and apply Bayes' Rule to real-world problems.
2. Develop and apply probabilistic models to solve real-world problems; Compare and contrast discrete and continuous probability distributions; Select and apply named probability distributions in a given context.

Standards for Block 2 – Data Wrangling & Visualization:

3. Describe and evaluate the context for a statistic, statistical test, or procedure, including identification of sampling biases and evaluation of data collection design; Propose improvements or alternatives when appropriate.
4. Compute, interpret, and compare measures of typical and spread; Evaluate the suitability of different measures given the data and context.
5. Create appropriate graphical summaries given the data and context; Evaluate the strengths and limitations of each type of graphical summary.

Standards for Block 3 – Hypothesis Testing:

6. Compare and contrast sample distributions and sampling distributions; Compare and contrast sampling and resampling; Compute and interpret standard error in context and explain its meaning to a non-expert.
7. Compute and interpret confidence intervals and explain their meaning to a non-expert; Articulate how confidence intervals relate to statistical inference.
8. State and apply the four steps of a hypothesis test in a given context; Relate hypothesis testing to confidence intervals; Articulate the limits and potential abuses of hypothesis testing.
9. Define and interpret Type I and Type II errors; Analyze their relationship and trade-offs in decision-making.

Standards for Block 4 – Linear Regression:

10. Specify and fit linear models; Interpret model coefficients and predicted values in a contextually meaningful way; Conduct and interpret inference for linear models; Evaluate linear model performance using appropriate statistical measures.
11. Evaluate the assumptions, limitations, and potential misuses of linear regression models; Assess model assumptions using graphical diagnostics; Explain phenomena such as Simpson’s paradox; Articulate risks associated with multiple testing and p-hacking; Communicate these issues to both expert and non-expert audiences (using accessible language and examples).
12. State and apply the Central Limit Theorem in varied contexts; Describe how the Central Limit Theorem relates to the sampling distribution.

5.e. Daily Course Rhythm

Each day, you are expected to read the assigned sections from our books and complete a reading guide. Then during class, we will review some of the big ideas from the readings and apply those ideas.

5.f. Workload

According to federal standards, each four-credit course should equate to at least 180 hours of work over the semester. If you are taking 16 credits, that equates to 720 hours of work over the 15 weeks of the semester, from the first day of classes until the end of the final exam period. In the case of this course, you will spend nearly 12 hours each week on my class alone (including our 4.5 hours of class meetings per week).¹

In considering the work for this course, I believe that the approximate 7.5 hours per week *outside* of class will breakdown something along these lines:

- 3 hours of class prep - One hour per lecture for reading the book and prep activities
- 1 hour per week on the project
- 2 hours per week on homework

¹Each of our 75 minute meetings is “counted” as 90 minutes. Since we meet 3 times per week, our “in-class contact time” is counted as 4.5 hours.

- 1.5 hours of “flex time” used to supplement any of the above areas and/or go to student hours

Notice that there are 1.5 hours of flex time. This is to accommodate weeks were you might want to spend more time on an assignment. For example, you might spend extra time studying in the weeks leading up to a graded review.

If you find that the time you are spending on this class is a lot more than 7.5 hours per week *outside* of class or a lot less, let’s check in.

5.f.i. *Spending time on You:*

The current system is designed to keep you working more than 40 hours per week just on your courses. This means that the hours not working on class work are precious and should be treated as such. This does not mean that every hour of every day should be filled with things deemed important by others, but more as a call to be intentional with your time. A few questions you might want to ask yourself include:

- Are you spending time on yourself and your wellbeing? What are you doing to restore yourself? How much time and space do these activities need to be effective?
- Are you giving to your courses the time that you want to be? Are you giving them the time during part of the day when your brain is ready to engage new ideas?
- Are you working doing your best thinking hours? If so, do you have any flexibility to adjust your daily schedule to be more present with your studies?
- Are you devoting time to being with your friends and/or your family? Are you present during those times?
- Are you losing hours to things or activities that are not high on your priority list? Alternatively, are you losing hours or days that you cannot recall what you did that day or why it so consumed you?

The adage tells us that “time is our most precious resource.” This statement frustrated me as a student because “I’ll have time after college/graduate school/post doc.” I have come to believe that this moment is the time that we have and if we are not intentional with our time today, right now, we will lose, not hours, not days, but months and years. I am working on a process of aligning my time to reflect my priorities as an educator, a researcher, a friend, and a family member. This is not a process that is completed in a few hours or a few days, but rather it is an iterative process of repeated and constant reflection.

6. RESOURCES SUPPORTING SDS 220

6.a. *Inclusivity for all students*

Smith is committed to providing support services and reasonable accommodations to all students with disabilities. Please inform me early in the term if there are aspects of the course that need to be modified to serve your learning, health, or person. You may speak with me after class, during student hours, or during an appointment. To request an accommodation, please register with the Accessibility Resource Center (ARC) at the beginning of the semester. To begin the process, either call 413-585-2071 or email arc@smith.edu.

Students in need of short-term academic advice or support can contact your class dean in the Dean of the College office. If you require some accommodation for religious or cultural purposes, please do not hesitate to let me know.

6.b. *Spinelli Center*

The Spinelli Center for Quantitative Learning is a resource to support students as they take STEM courses and as they engage in other quantitative work. To see the list of available peer tutors and other resources available at the Spinelli Center, please consult their [website](#).

6.c. *Jacobson Center*

Smith has an additional resource for writing support: the Jacobson Center for Writing, Teaching & Learning make an appointment to take your work to the Jacobson Center on [their website](#).

6.d. *SDS TAs*

The SDS program partners with the Spinelli Center to hire and train **excellent** TAs to support your learning. The TA hours are a great place to plan to meet up with others to work on course work.

The SDS TAs are in Sabin-Reed 301 during the following days:

- Sundays 7-9 PM
- Mondays 7-9 PM
- Tuesdays 7-9 PM
- Wednesdays 7-9 PM
- Thursdays 7-9 PM

6.e. *Library*

The library is a huge resource and the librarians are eager to help you with this (or any other) class!

Our books are available as E-books and on the web:

1. [OpenIntro Statistics, 4th edition](#) by David Diez, Mine Çetinkaya-Rundel, and Christopher D. Barr.
2. [Introduction to Modern Statistics, 2nd edition](#) by Mine Çetinkaya-Rundel and Johanna Hardin.
3. [Modern Dive, 2nd edition](#) by Chester Ismay, Albert Y. Kim, and Arturo Valdivia.

Note

Please note that for anything denoted as an E-Book at the library, there are limits placed on how many people can view the source. Instead, you can download sections (or whole books) for a period of time directly from the library.

6.f. Student Hours

Student hours are **drop-in hours**. These hours are for *you* to meet with me about anything! They are times on my calendar with *nothing else* on them.

I encourage you to add student hours to your weekly calendar. Below is an incomplete list of great reasons to hangout in gather before or after class or to make an appointment:

- You haven't had the chance to chat with me yet
- You want to see what my office looks like
- You have a question on a reading, assignment, or activity
- You want to color in a butterfly drawing
- You want to share about the course, your time at Smith, or about yourself
- You read something about machine learning, computer science, statistics, or math that you have questions about
- You are thinking about what you want to do after Smith

Note

Student hours will be:

- Mondays from 10am to 11:30am
- Thursdays from 2:30pm to 4pm, and
- By [appointment](#)

6.f.i. Appointments:

You are welcome and encouraged to make individual appointments with me. To make an appointment, please check my [appointment calendar](#) for appointment slots that are set aside each week.

Note

To book an appointment with me, go to my [appointment calendar](#)

7. ACADEMIC INTEGRITY

Nothing is held more dear to a researcher than their integrity. To researchers, the most precious commodity is our ideas. While some researchers build tangible items that can be protected by patents, many researchers' products are ideas and theories, which are less likely to be protected under things like patents. Instead, it is academic integrity that governs our research communities, protecting our ideas from being stolen. The system of academic integrity relies on researchers trusting each other to be honest and to act with integrity.

7.a. *Academic Integrity at Smith*

As you may be aware of, Smith is continuing to transition towards an Academic Integrity Statement. This comes out of Smith's tradition of an Honor Code. This tradition goes beyond inviting students to act with integrity, instead it welcomes students as equal participants of the learning community, imbuing students with the same level of trust that we extend to our collaborators and colleagues.

The Smith College Academic Integrity Statement (as stated on the [Academic Integrity page](#)) says:

Smith College provides its students with a world-class liberal arts education. The purpose of this education is not only to provide students opportunities for intellectual growth and advancement, but also to prepare its graduates to make powerful contributions to the world. Upholding the integrity of a Smith education, then, is at once a responsibility to oneself and to the community.

The Academic Integrity Board (AIB) defines academic integrity as the alignment of students' behaviors in academic courses with Smith's commitment to the honest pursuit of genuine learning. Smith students are responsible for upholding their own integrity by adhering to all course policies and properly acknowledging all sources used in preparing academic work. When assignments require students to submit work that is the product of their own intellectual labor, faculty expect that students have neither used unauthorized resources nor engaged in unauthorized collaboration with others. When courses require students to submit work that is the product of intellectual engagement with fellow students, students should follow all of the guidelines set out for collaboration. All submitted coursework of any kind must be the original work of the student(s). Faculty are expected to clearly communicate to students how honest engagement is defined in each course.

According to Smith's Statement of Purpose, "the world needs a place where knowledge is not the end, but merely the beginning of creating incalculable good. And we will always be that place." Delivering on this promise demands that students and faculty hold each other to the highest standard of academic integrity.

This trust bestowed to students in this statement is the same trust that exists in and among researchers within a research community, the same trust that exists between me and my collaborators. Simply put, at Smith, under the Academic Integrity Statement, I trust that you will each act with integrity, citing sources when you celebrate others' ideas and noting who you work with when collaborating.

This trust manifests in how assignments are created, in how work is completed, and in how we resolve instances where the trust has been broken. I create assignments knowing that while there are resources online that can offer complete solutions, and I trust you to not seek out such complete resources. I trust that if you do stumble on a solution guide, you will not use it. Under this Academic Integrity Statement, students are expected to keep careful notes about the resources that they consult and the people that they collaborate with. We get to assume that the work handed in by a student is the creation of that student. Lastly, when there are concerns regarding potential violations to Academic Integrity, the resolution is determined by AIB, which has established by the community and trusted to seek restoration of the whole community's trust through education and action.

I regard our Academic Integrity with deep and profound respect. Being an educator at Smith with this Academic Integrity Statement means we—your faculty—begin from a place of trust without any underlying suspicion of our students. Simply put, we—instructors and students—work from the assumption that all are acting in good faith and with the utmost integrity. This is an assumption that cannot be made at a school without this tradition and it is why it is a privilege to be both an instructor and a student at Smith under this Academic Integrity Statement.

7.b. Academic Integrity Practicalities

The work you submit should be your own and created by you, unless explicitly listed as a group assignment. With the exception of the starred problems, I do encourage you to ask for help from your peers or myself when you have questions; however, copying is never allowed. The line between copying and helping is subtle. Below are a few guidelines:

- Do not share nor give your work with other students; instead, offer to discuss the big ideas of the task at hand.
- Do not look at someone else's work (including online solutions); instead, ask if you could talk with them about your ideas and share where you are getting stuck.
- Acknowledge those you talk or work with at the top of **every** assignment, **including any generative AI tools**. I will not dock points for getting acknowledged help from others. If you generate a solution or an argument with someone, in addition to acknowledging that person, recreate the solution or argument on your own in your own words.

If you think you may have crossed the line between helping and copying, please talk to me. Do not let me discover that the line was crossed. If a violation of Academic Integrity is suspected, the student will be informed and will be given the opportunity

to meet with the instructor. After the meeting, the instructor will decide if further consideration by the AIB is merited, and if so, the suspected violation will be reported to the AIB by the instructor.

If you are unsure about how Smith's Academic Integrity Statement applies to our course, please consult with me or a class dean. We are happy to discuss this with you.

7.c. Creation vs. Curation

In my role as your instructor, I am **curating** resources for you that I feel are best for the teaching and learning of statistics. In this process of curation, I will give credit to those whose materials I have used and I will only use materials as allowable by copyright and fair use.

In this course, you will be asked to **create** solutions to problem sets, lab reports, and a final project. In these acts of creation, you are demonstrating your most current understanding of the concepts in our course. This course will give you lots of chances to demonstrate what you know and lots of flexibility to repeatedly try again.

I am aware that there is a tension between the curation that I am modeling and the creating that is being asked of you. My goal is to provide the best resources to you, which requires reading and sorting through many sources (curation). Your goal is to learn as much statistics as deeply as possible, which requires a lot of practice, trying out statistical ideas, reflecting on your attempts, and trying again (creation).

8. ASSESSMENT

This course will use a version of specifications-based grading. In this course, there are 28 tokens to collect: 12 Statistics Tokens from the graded reviews (GRs), 9 Application Tokens, and 7 Professional Development tokens. The token breakdown is as follows

- 15 Statistics Tokens: GRs (with each token tied to a particular skill)
- 9 Application Tokens:
 - 2 Tokens: Statistical Specialty
 - 3 Tokens: Semester-Long Group Project
 - 4 Tokens: Individual Homework Assignments
- 7 Professional Development Tokens:
 - 2 Tokens: Pre-Class Reading
 - 2 Tokens: Course Participation
 - 3 Tokens: Community Engagement

8.a. Graded Reviews

There are four graded reviews (GRs) throughout the term plus one Final GR during the finals period. These are just moments in time to share what your current understanding of the course material is. There are 12 tokens, each representing core skills for the course. During each GR, you can attempt as many or as few tokens as you would like. Your goal, by the end of the semester, is to collect all 12 tokens. (Think of this like Pokemon, but for statistics)

The first GR will have the first 2 tokens available, the second will have the first 5, the third GR will have the first 9, and fourth GR will have all 12 tokens. The final will also have all 12 tokens.

Put another way:

- Tokens 1-2 will appear on GRs 1-4 plus the final
- Tokens 3-5 will appear on GRs 2-4 plus the final
- Tokens 6-9 will appear on GRs 3-4 plus the final
- Tokens 10-12 will appear on GR 4 plus the final

You do not need to (nor should you) attempt all tokens on each GR, but you do need to collect all 12 tokens by the end of the term.

8.a.i. Levels for Statistics Tokens on GRs:

There are five levels for each statistics token:

- P - Platinum: The student understands the material deeply enough to teach it to others. This represents a different level of proficiency – the ability to explain, critique, and guide others.
- G – Gold: The student understands the material deeply enough to apply and interpret the concepts independently and correctly, with only minor or infrequent communication errors

- S – Silver: The student understands the material well enough to apply and interpret the concepts independently most of the time, with occasional minor justification errors.
- B – Bronze: The student demonstrates partial understanding of the material; conceptual gaps, reasoning errors, or major misconceptions are present and must be addressed before the material can be applied reliably.
- CBA – cannot be assessed: The student has provided incoherent or incomplete evidence of understanding.

Note these level descriptions match to the current version of DataSci 220 at USAFA.

8.a.ii. *Achieving Statistics Tokens on GRs:*

The 12 tokens associated to the GRs are tied to the 12 statistics tokens. The most recent level for each token is the recorded level. This means that if you earn a platinum level for token 1 on the first GR, then you never have to attempt this token again. But, if you earn silver for token 4 on the first challenge and then a bronze for token 4 on the second challenge, your current level for token 4 is bronze.

Note

You do not need to attempt all tokens on each GR. This means that you get to pick and choose which skills you are ready for on each GR.

For example, you may elect to just study token 1 for the first GR, then 2-4 for the second one, 5-8 for the third, and then 9-11 for the fourth one. For the final, you will have to attempt 12 plus any tokens you want to try again on the final.

Alternatively, you may elect to study both tokens for the first GR to see how the system works, and then study 3-5 plus one from the first GR that didn't go as well. Then for the third one, you might again study three new ones (say 6-8) plus one or two that you want to “polish” from the first five tokens that you studied for the first two GRs.

Warning

You need to collect all 12 tokens by the end of the term. This means that you will need to start to think a bit more strategically as you prepare for each GR

Note

You **only** need to take the final GR if you *want* to “upgrade” any token levels.

8.b. *Application Tokens*

The 9 application tokens are attached to course activities where you apply your statistical knowledge. For the application tokens, there are four levels for each token: Gold, Silver, Bronze, CBA (can't be assessed)

8.b.i. *Statistical Specialty: 2 Tokens:*

In the first half of the semester, you will be asked to select one statistical specialty: 1) communication and visualization, 2) mathematical foundations, 3) computational methods, or 4) statistical extensions. Each of these four specialties have two standards associated with them. You will complete a reading guide and assignment for each standard in your chosen specialty.

8.b.ii. *Group Project: 3 Tokens:*

The tokens associated to the group project will be tied to rubrics. Like the SDS 100 project, there will be individual and group tasks. The grading rubrics will be introduced with the project.

8.b.iii. *Individual Homework Assignments: 4 Tokens:*

There are 8 homework assignments over the course. Each homework has two parts and are each worth 50 points. There are two homeworks per block that together represent one token. To earn a Gold for one homework token, you need to earn 80 or more points (of the 100 total). To earn a silver, you need to earn 60 or more, and to earn a Bronze, you need to earn 40 or more points.

Note

Handing in incomplete homework is better than handing in nothing!

Keep in mind that there are two parts to each assignment: Concept Checks and Core Competencies. The Concept Check questions can be done immediately after each class.

8.c. *Professional Development Tokens*

The 7 professional tokens are attached to course activities where you develop the habits associated with developing as a statistician and as a professional.

8.c.i. *Pre-Class Reading: 2 Tokens:*

During the first three blocks of the course, you will be given a **printed** reading guide to complete for each lesson. The two tokens associated to the pre-class readings are based on your good-faith completion of these reading guides.

8.c.ii. *Engagement & Class Preparation: 2 Tokens:*

Two of these habit tokens will be earned with student input. In the sixth week of the course and during the last week of classes, you will be asked to complete a self-evaluation of your class engagement in context of the class engagement list. Your self-evaluations will be a critical part of determining the levels associated to these three tokens, but not the only determining factor.

8.c.iii. *Community Engagement: 3 Tokens:*

As a placed-based learning community, there are a number of co-curricular and extra-curricular events at Smith each week. As part of your work for this course, you are expected to engage in the out-of-class learning opportunities.

The expectation is that you will attend 4 events this semester. Pre-approved eligible events include:

- Any SDS or CSC Seminar
- The SDS Presentation of the Major
- The CSC Presentation of the Major
- Any Event that the instructor advertises on our slack (with the tag “community-engagement”)
- At most one student-led event, such as a sporting event or a student-org event

If there is an event that you would like to attend towards this expectation that is not listed above, please reach out to me on slack.

In early Decemeber, you will be asked to reflect on your community engagement as it relates to you growth as a statistician and as a learner. Your tokens are assigned based off: attendance at your four events (2 tokens) and the quality of your reflection (1 token).

8.d. *Flexibility Systems*

This course has a number of flexibility systems built in:

- Notice that for homework and reading guides you only need to complete about 80% of the work in each division to earn a top score for that division.
- For the statistics tokens on the GRs, you have between 2 and 5 attempts for each token. This means that you have time to iteratively study, attempt, reflect, and reattempt for each token.

8.d.i. *Latework:*

All work is due at 10am on the date indicated on the Detailed Course Schedule. This is a great moment to note the due dates of all assignments in the system (ie. planner, calendar, etc) that you use to keep track of your work. (If you want to really stretch here, do this for all major assignments in all your courses.)

In general, work should not be late. Later course content builds on previous content, and if one is spending time on an assignment that is passed due, then there is less time for the next assignment. Additionally the grading for this course has a number of flexibility systems in place, and I encourage that you take advantage of them as needed.

This all being said, if you find yourself in need of a bit more time, send me a brief slack message at least a day **before** the work is due. In your message include 1) what flexibility systems you’ve already used (see above), 2) why it is important to your learning that you get an extension, 3) propose a new due date for yourself, and 4)

why the proposed extension will not hinder your ability to get the next assignment done on time. Your message should be *no more than seven sentences long*.

8.d.ii. *Missing a GR:*

The self-schedule GR windows are listed on the Detailed Course Schedule. That being said, things do come up and you may be in a position where you need to miss a GR. Under this grading system, you are working to demonstrate your most complete understanding of the course material by the end of the term. So you can miss a GR for any reason knowing that you can attempt the tokens from the missed challenge on a later GR.

Warning

Missing two or more GRs is not advisable.

This being said, if something comes up, I would rather you reach out and check in, than not reach out.

8.e. *Turning Tokens into Letter Grades*

This course uses a contract system, in that your instructor will **guarantee** you grades based off the following process. This system is informed by previous iterations of this version of the course both here at Smith and at USAFA.

Here's how it works:

- For the contract grading, Platinums and Golds are treated as the same. This means that you are **never** required to earn Platinums.
- Your final grade is determined with a base GPA score and then adjustments
- The following guarantees *may* be lowered but they will *not* be raised.

The following table shows the guaranteed minimum final grade corresponding to the GPA points:

| GPA points | 4.0 | 3.0 | 2.0 | 1.0 | |Guaranteed Grade| A | B | C | D |

The base GPA score will be determined based on the statistical tokens you earn through the GRs:

Letter Grade	Requirement
4.0	At least 8 Golds, remainder Silver and Bronze with a max of 1 Bronze
3.0	At least 6 Golds, remainder Silver and Bronze with a max of 3 Bronze
2.0	At least 6 Golds or Silvers, remainder Bronze with no CBA's
1.0	At least 4 Golds or Silvers, remainder Bronze with no more than 2 CBA's

Note

Higher levels can count for lower level totals, but not vice versa. For example, a silver level can count towards your bronze total but a silver level cannot count towards your gold total.

From your base GPA points, we then apply an adjustment based on the tokens you earn in applications (9 available) and in professional development (7 available). Additionally, earning of Platinum tokens on GRs will also add to your final grade adjustment.

| Adjustment | Platinum | Application Tokens | Professional Development |

to Base Score	GR tokens	(9 Available)	(7 Available)
+0.67	7 or more Ps		
+0.5	5 or more Ps	6 Golds remainder Silvers	7 Golds
+0.33	3 or more Ps	4 Golds remainder Silvers	6 Golds, no CBAs
+0.0		2 Golds remainder Silvers or Bronze	5 Golds, no CBAs
-0.16		2 Golds remainder Silvers or Bronze max of 1 CBA	3 Golds, no CBAs
-0.33		6 Silvers max of 3 CBAs	5 Silvers, max of 1 CBA
• 0.5		Does not meet above requirements	Does not meet above requirements

8.e.i. *Example::*

Suppose that Sophia earns the following:

- GR tokens (12 in total): 1 P, 6 Gold, 3 Silver, and 2 Bronze
- Application Tokens (9 total): 6 Gold and 3 Silvers
- Professional Development (7 total): 5 Golds, 1 Silver, and 1 Bronze

Sophia's base score is 3.0 because she has 7 total between Gold and Platinum and she has less than 3 Bronzes. Then her adjustments are computed as follows:

- +0.0 for her one platinum
- +0.5 for her application adjustment
- +0.16 for her professional development adjustment

This means that her guaranteed score is:

At the instructor's discretion, these grade cutoffs may be lowered at the end of the semester, but they will never be raised. Cut-offs not stated here for the pluses and minuses will be determined at the end of the semester.